

Not Every Breakdown Is Quantum: A Gaussian-Pulse Benchmark for Landau–Lifshitz Accuracy

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ABSTRACT: Discussions of radiation reaction often use “Landau–Lifshitz breakdown” to mean two different things: failure of a classical reduction of order, and failure of any classical emission model once quantum recoil becomes important. This paper separates those questions in the nonrelativistic one-dimensional Abraham–Lorentz limit of the Lorentz–Abraham–Dirac equation. For an electron driven by a spatially uniform Gaussian electric pulse, the physical non-runaway acceleration is an exact future-weighted integral of the force, while the Landau–Lifshitz acceleration is its first local reduction-of-order approximation. Direct quadrature and a Gaussian-moment expansion show that the normalized root-mean-square error obeys $D_{\text{rms}} = (\sqrt{3}/2)\epsilon^2[1 - \epsilon^2 + O(\epsilon^4)]$, with $\epsilon = \tau_e/T$, and that the maximum error scales as $D_{\text{max}} \sim \epsilon^2$ for smooth pulses. The numerical scan is then recast as a diagnostic map in the (ϵ, χ_e) plane: vertical contours encode the computed classical truncation error, while horizontal guides mark the quantum-recoil and stochastic-emission regimes. In this linear benchmark, changing field amplitude or beam energy changes χ_e but not the normalized LL–LAD error. Since femtosecond laboratory pulses have $\epsilon \sim 10^{-9}$, this source of LL error is negligible in strong-field experiments; discrepancies with data should instead be interpreted through quantum emission, beam and field modelling, or detector-background effects.

KEYWORDS: Radiation reaction; Landau–Lifshitz equation; Lorentz–Abraham–Dirac equation; Quantum nonlinearity parameter; Strong-field QED; Numerical benchmark

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“The first principle is that you must not fool yourself.”

— Richard P. Feynman, *Cargo Cult Science*

1 Introduction

Radiation reaction is a basic but subtle problem in classical electrodynamics. An accelerating charge radiates energy and momentum, so a complete equation of motion must include the recoil of the charge on its own field. Imposing energy–momentum conservation on a radiating point charge leads to the Lorentz–Abraham–Dirac (LAD) equation, the covariant point-particle form of this idea. However, LAD is difficult to use directly: it is third order in time and admits runaway solutions whose acceleration grows without bound even after the external force is switched off [1, 3]. The Landau–Lifshitz (LL) equation removes this pathology by reducing the order of LAD and replacing the self-force with a perturbative expression built from the external field and the Lorentz-force motion [2, 5]. For this reason, LL is the form most commonly used in practical calculations.

The difficulty is that the phrase “LL breaks down” can refer to two different questions. The first is a classical approximation question: does LL still track the physical, non-runaway branch of the LAD equation? The second is a physical-validity question: can any deterministic classical equation describe radiation reaction once photon recoil and the stochastic character of emission become important? Strong-field laser physics often couples these two questions, but it does not make them identical. The quantum nonlinearity parameter

$$\chi_e \simeq \frac{\gamma E_{\text{eff}}}{E_S},$$

where γ is the electron Lorentz factor and E_{eff} is the external field it experiences, compares the electron rest-frame field with the Schwinger field E_S . It is the standard marker for the onset of quantum emission effects [4]. Recent high-intensity experiments are therefore interpreted through classical, quantum-continuous, and quantum-stochastic models of radiation reaction. The collision experiments of Cole et al. and Poder et al. reported signatures of radiation reaction consistent with quantum descriptions [12, 13], and a recent measurement reports a high-significance observation favouring quantum models over the classical one in the strong-field regime [14]. This vocabulary makes it tempting to identify large χ_e with the failure of LL itself. The present paper argues that this identification hides a simpler, purely classical issue underneath the quantum one.

This paper separates the two notions of breakdown in the simplest setting where the LAD–LL comparison can be written explicitly: the nonrelativistic one-dimensional Abraham–Lorentz limit of LAD, for an electron driven by a spatially uniform Gaussian electric pulse. In this limit the physical non-runaway acceleration does not require unstable third-order integration. It is given in closed form by a future-weighted integral of the applied force, and the LL acceleration is exactly its first local reduction-of-order approximation. The comparison then depends on a single dimensionless ratio, the pulse-variation parameter

$$\epsilon = \frac{\tau_e}{T},$$

with τ_e the electron radiation-reaction time and T the characteristic pulse duration.

The research question is therefore the following. In the nonrelativistic one-dimensional Abraham–Lorentz limit of the LAD equation for an electron driven by a smooth Gaussian electric pulse, how does the LL reduction-of-order approximation deviate from the exact non-runaway acceleration as a function of $\epsilon = \tau_e/T$, and how can this classical accuracy criterion be placed alongside χ_e to distinguish reduction-of-order breakdown from quantum radiation-reaction onset?

The claim attached to this question is deliberately modest. This paper proposes no new equation of motion and does not attempt a strong-field QED calculation. Earlier pedagogical work has already compared Abraham–Lorentz and Landau–Lifshitz dynamics, including finite-size-charge tests and simple force profiles [6, 7]. The present contribution is narrower but more specific: it turns the Gaussian force profile into a calibrated diagnostic benchmark with normalized D_{rms} and D_{max} metrics, an analytic coefficient $D_{\text{rms}}/\epsilon^2 \rightarrow \sqrt{3}/2$ with its first correction, and a data-backed error map that separates the classical parameter ϵ from χ_e .

The distinction is not merely a matter of terminology. Particle-in-cell and single-particle codes used to model laser–plasma interaction and next-generation accelerator stages must choose between a classical radiation-reaction force, a reduced-order equation such as LL, and a stochastic quantum-emission module. This choice is often justified by quoting a value of χ_e [4, 5]. Separating the reduction-of-order error from the quantum-validity question clarifies which choice a given parameter regime actually supports, and which question remains open. In this sense the work is best read as a reproducible simulation-methods benchmark in JINST format: it supplies a small, exactly checkable test problem for a modelling distinction that appears in larger strong-field codes.

This separation also fixes the scope of the paper. Because the exact non-runaway acceleration is available in closed integral form, the mathematical core remains fully classical and analytically controlled. The strong-field material enters only as interpretation, not as technical overhead. The treatment is nonrelativistic, one-dimensional, and driven by an externally prescribed field; large values of ϵ are used as a controlled diagnostic of the approximation, not as a claim about realistic laboratory pulses, since $\tau_e \sim 6 \times 10^{-24}$ s makes $\epsilon \sim 1$ physically extreme. The project therefore remains within the scope of an undergraduate electrodynamics paper while still connecting to the contemporary literature, where LL, LAD, and quantum-emission models are often compared in the same experimental setting.

The paper is organized as follows. Section 2 reviews the LAD equation, its physical non-runaway branch, and the LL reduction of order. Section 3 constructs the exact non-runaway benchmark in the one-dimensional Abraham–Lorentz limit. Section 4 derives the expected LL error scaling. Section 5 applies the comparison to a Gaussian pulse and reports a compact numerical illustration. Section 6 then interprets the result in relation to χ_e and strong-field radiation reaction.

2 Classical radiation-reaction background

This section fixes the three objects compared in the rest of the paper: the LAD equation, the physical non-runaway branch selected from its solutions, and the LL equation understood as a reduction of order. The aim is to make precise what “LL approximates LAD” means before introducing any pulse shape or numerical calculation. None of this background is new; it is included so that the toy model in Section 3 measures the correct object.

2.1 The LAD equation and its pathologies

In covariant notation, with four-velocity u^μ , proper time τ , and metric signature $(+---)$, the LAD equation can be written schematically as

$$m \frac{du^\mu}{d\tau} = e F^{\mu\nu} u_\nu + m \tau_e \left(\frac{d^2 u^\mu}{d\tau^2} + \frac{1}{c^2} \frac{du^\alpha}{d\tau} \frac{du_\alpha}{d\tau} u^\mu \right), \quad (2.1)$$

where the radiation-reaction coupling is set by the characteristic time

$$\tau_e = \frac{2}{3} \frac{e^2}{4\pi\epsilon_0 mc^3} \simeq 6.26 \times 10^{-24} \text{ s} \quad (2.2)$$

for the electron. Numerically, τ_e is the time light takes to cross about two-thirds of the classical electron radius. Its extreme smallness is why the radiation-reaction term is a tiny correction in almost all laboratory situations. The two terms in the bracket play different roles. The quadratic piece, proportional to $(du^\alpha/d\tau)(du_\alpha/d\tau)u^\mu$, is the dissipative part: it carries away four-momentum at the relativistic Larmor rate and has the character of radiative friction. The Schott term $d^2u^\mu/d\tau^2$ is the source of the mathematical difficulty. Because it contains a second proper-time derivative of the velocity, equation (2.1) is third order in the position. Position and velocity at one instant no longer determine the trajectory; an initial acceleration must also be specified.

This extra initial datum is where the pathology enters. The freedom it represents is a runaway mode: even with $F^{\mu\nu} = 0$, the equation admits self-accelerating histories in which the proper acceleration grows like e^{τ/τ_e} [1, 3, 5]. Generic initial accelerations excite this mode, and the particle accelerates itself to the speed of light with no applied force. These histories are genuine mathematical solutions of the differential equation, but they are not accepted as physical electron motions. A prescription is therefore needed to discard them.

2.2 The physical non-runaway branch

Removing the runaways requires an additional condition. The standard condition, due to Dirac, is imposed in the future rather than the past: the acceleration is required to vanish asymptotically once the external force is switched off. Tuning the initial acceleration to meet this condition selects a distinguished branch of solutions and removes the runaway freedom, at the cost of making the dynamics nonlocal in time. On this branch the acceleration at a given instant depends on the force still to come, and the charge can begin to accelerate slightly before the force arrives, over a window of order τ_e . This pre-acceleration is a property of the selected branch, not a numerical artifact of a particular integration scheme, and at accessible pulse durations it is far too small and too brief to be observed.

Spohn’s critical-manifold picture gives this selection a geometric interpretation. The physically admissible solutions lie on a low-dimensional invariant (“critical”) manifold in the LAD phase space. Transverse to that manifold, the dynamics is the runaway: off-manifold perturbations grow like e^{τ/τ_e} , so the transverse directions are repelling in forward time. Dirac’s asymptotic condition is what restricts the motion to the manifold. The reduced motion on the manifold can then be obtained perturbatively, which is the modern justification for replacing the full third-order system with an effective second-order equation [3]. The same physical branch can be studied by other routes; for planar motion in a constant field, for instance, the dynamics reduces to a single second-order equation whose large-proper-time asymptotics can be obtained explicitly [20]. The formal structure of the runaway-free reduction, including the way the perturbative series encodes the discarded modes, has also been analysed using transseries [10].

The practical consequence is important for the rest of the paper. The correct benchmark for LL is the physical branch, not arbitrary LAD initial data; comparing LL with a runaway-contaminated solution would test the wrong object. It also means that the benchmark can be built from a stable

closed-form representation of the non-runaway solution, as done in Section 3, rather than from an unstable third-order initial-value integration.

2.3 LL as a reduction of order

The full relativistic LL equation replaces the self-force in (2.1) with terms built from derivatives of the external field along the trajectory and from quadratic field combinations [2]. The construction is a reduction of order: one takes the small radiation-reaction term, which carries the troublesome higher derivative, and substitutes into it the lowest-order (Lorentz-force) acceleration, $du^\mu/d\tau \approx (e/m)F^{\mu\nu}u_\nu$. Differentiating that lowest-order relation turns the Schott term into an expression involving the field gradient and the field squared, with no second derivative of the velocity left to supply as initial data. The result is a second-order equation with no runaways, and it is the form normally adopted as the working classical model of radiation reaction [5].

Two cautions follow from this step and shape the present project. First, reduction of order is a perturbative substitution. It is valid when the self-force is a genuinely small correction, so it should be read as a controlled local approximation for smooth fields, not as an exact rewriting of the physical branch. Second, the reduction-of-order expansion is generally asymptotic rather than convergent: LL is its first term, often an optimal low-order truncation, but additional terms need not improve it and can miss short-time, resummed behaviour [9]. Both cautions are easiest to see in the simplest case, where the tensor structure of (2.1) collapses and the substitution can be carried out by hand. In the nonrelativistic one-dimensional limit, the radiation-reaction term $\tau_e \dot{a}$ has the lowest-order acceleration $a \approx F/m$ inserted into it, giving the local approximation $a_{\text{LL}} = F/m + \tau_e \dot{F}/m$. The next section therefore specializes to this Abraham–Lorentz limit, in which the physical branch is an elementary integral and LL is its first derivative correction.

3 Exact non-runaway benchmark in the Abraham–Lorentz limit

3.1 Nonrelativistic reduction

Specializing the covariant equation (2.1) to one spatial dimension and nonrelativistic motion reduces its tensor structure to a single scalar relation. For an electron in a spatially uniform, time-dependent electric field $E(t)$, the Lorentz factor is $\gamma \approx 1$, the magnetic force vanishes, and the dissipative quadratic term in (2.1) is higher order in v/c and drops at leading order. The self-force then keeps only the Schott contribution, and the equation of motion becomes the Abraham–Lorentz form

$$a(t) = \frac{F(t)}{m} + \tau_e \dot{a}(t), \quad (3.1)$$

where $F(t) = -eE(t)$ is the externally prescribed force and $a = \dot{v}$. The field is taken as a function of time alone: spatial gradients are dropped, so the force does not depend on the particle position and the electron does not act back on the field. This is the standard test-charge reading of the problem, and it is the price of solvability; its limits are revisited in Section 6. Equation (3.1) is the nonrelativistic one-dimensional Abraham–Lorentz limit of LAD, and it inherits both difficulties identified in Section 2. It is effectively of higher order, because the self-force depends on $\dot{a}$ rather than on a alone, and its homogeneous solutions run away.

Setting $F = 0$ in equation (3.1) makes this explicit: the source-free equation $a = \tau_e \dot{a}$ has the solution $a(t) \propto e^{t/\tau_e}$. Every solution of the driven equation is therefore fixed only up to an additive multiple of this runaway mode, so equation (3.1) by itself does not determine the physical motion. Selecting the physical branch requires the supplementary condition discussed in Section 2.2, now applied to the scalar equation: the acceleration must not grow like the runaway mode once the pulse has passed.

3.2 Exact non-runaway solution

Equation (3.1) is a linear first-order equation for $a(t)$. Rearranging it as $\dot{a} - a/\tau_e = -F/(m\tau_e)$ and multiplying by the integrating factor e^{-t/τ_e} gives

$$\frac{d}{dt} [a(t) e^{-t/\tau_e}] = -\frac{1}{m\tau_e} F(t) e^{-t/\tau_e}. \quad (3.2)$$

The general solution is a particular integral plus a multiple of the homogeneous runaway e^{t/τ_e} . Integrating equation (3.2) from t to ∞ and imposing the non-runaway condition $a(t) e^{-t/\tau_e} \rightarrow 0$ as $t \rightarrow \infty$ removes the boundary term at infinity, fixes the homogeneous coefficient to zero, and yields a unique result,

$$a_{\text{LAD}}(t) = \frac{1}{m\tau_e} \int_t^\infty \exp\left[-\frac{t' - t}{\tau_e}\right] F(t') dt'. \quad (3.3)$$

Equation (3.3) is the exact physical acceleration in this model and the benchmark against which the LL approximation is measured throughout the paper. The substitution $t' = t + \tau_e u$ casts it in the compact form

$$a_{\text{LAD}}(t) = \frac{1}{m} \int_0^\infty e^{-u} F(t + \tau_e u) du, \quad (3.4)$$

which is the representation expanded in Section 4 and evaluated numerically in Section 5.

Equation (3.3) can also be read as a Green-function statement. The operator on the left of equation (3.1), written $1 - \tau_e d/dt$ acting on a , has a future-decaying inverse whose kernel is $\tau_e^{-1} e^{-(t'-t)/\tau_e}$ for $t' > t$ and zero otherwise. Equation (3.3) is the convolution of that kernel with F/m . Selecting the future-decaying Green function rather than the past-decaying one is the same choice as discarding the runaway mode, now phrased as a boundary condition for an inhomogeneous linear equation. This is the one-dimensional, nonrelativistic image of the critical-manifold reduction of Section 2.2: the manifold is the single physical solution (3.3), and the discarded transverse direction is the growing exponential.

Two properties of equation (3.3) carry the physics. The acceleration at time t is a weighted average of the force at *later* times $t' > t$, with the exponential kernel $e^{-(t'-t)/\tau_e}$ setting the weight. The electron therefore responds to the force over a forward window of width of order τ_e . This forward weighting is the origin of pre-acceleration: the particle starts to move slightly before the force reaches it. Because the future condition was imposed at $t \rightarrow +\infty$, the solution is causal in the operational sense that no information about times earlier than the relevant force is required; the apparent acausality of pre-acceleration is confined to the radiation-reaction timescale τ_e . The kernel is normalized, $\frac{1}{\tau_e} \int_t^\infty e^{-(t'-t)/\tau_e} dt' = 1$, so a static force returns $a = F/m$ and the ordinary Newtonian result is recovered whenever radiation reaction is negligible.

3.3 Meaning of the benchmark

The usefulness of equation (3.3) is both practical and conceptual. It writes the non-runaway branch as a convergent integral of the applied force, so the physical acceleration can be evaluated by quadrature without integrating the unstable third-order dynamics forward in time. This makes a clean numerical comparison with LL possible, and it is why Section 5 computes equation (3.3) directly rather than solving equation (3.1) as an initial-value problem. The exponential suppression in equation (3.4) also makes the quadrature simple: the integrand decays like e^{-u} , so a modest truncation of the u integral captures the solution to high accuracy.

The construction is exact only within the simplified model: the field is uniform and prescribed, the motion is nonrelativistic and one-dimensional, and the point-particle picture is assumed. Inside those limits the integral is not an approximation but the physical branch itself. This identification is consistent with recent analysis treating LL as the first term of a reduction-of-order series whose higher orders form a divergent, asymptotic expansion, and showing in solvable field configurations that resumming the series reconstructs a runaway-free physical solution rather than a new equation [9]. Equation (3.3) is the nonrelativistic, closed-form instance of that physical branch. It is also the natural object against which to test how short a pulse may be, in units of τ_e , before the first local correction stops being adequate; that question becomes quantitative once the kernel acts on a specific force profile. The next section expands equation (3.4) to recover LL and to identify the leading discrepancy.

4 LL approximation and expected deviation

4.1 Expansion of the physical solution

The compact form (3.4) of the physical acceleration is built for expansion. When the force is smooth and varies slowly on the scale τ_e , the shift $\tau_e u$ inside $F(t + \tau_e u)$ is small over the range where the kernel e^{-u} has support. We can then Taylor-expand F about t ,

$$F(t + \tau_e u) = \sum_{n=0}^{\infty} \frac{(\tau_e u)^n}{n!} F^{(n)}(t). \quad (4.1)$$

Inserting equation (4.1) into (3.4) and using the elementary moments of the exponential kernel,

$$\int_0^{\infty} e^{-u} u^n du = n!, \quad (4.2)$$

makes every term collapse to a single derivative, because the factorials cancel. The result is the local derivative series, to be read as an asymptotic expansion in τ_e (see Section 4.2),

$$a_{\text{LAD}}(t) \sim \frac{1}{m} \sum_{n=0}^{\infty} \tau_e^n F^{(n)}(t) = \frac{1}{m} [F(t) + \tau_e \dot{F}(t) + \tau_e^2 \ddot{F}(t) + \cdots]. \quad (4.3)$$

Each derivative of the force is accompanied by one extra power of τ_e , so the size of successive terms is controlled by how rapidly F varies relative to τ_e .

4.2 LL as the first local approximation

The nonrelativistic LL acceleration is the truncation of equation (4.3) after the first correction,

$$a_{\text{LL}}(t) = \frac{1}{m} [F(t) + \tau_e \dot{F}(t)], \quad (4.4)$$

which is exactly the reduction-of-order result obtained in Section 2.3 by substituting the lowest-order acceleration into the radiation-reaction term. Subtracting it from the exact series isolates the leading discrepancy,

$$a_{\text{LAD}}(t) - a_{\text{LL}}(t) = \frac{\tau_e^2}{m} \ddot{F}(t) + O(\tau_e^3). \quad (4.5)$$

If the force varies on a characteristic time scale T , then $F^{(n)} \sim F/T^n$, and the n th term in equation (4.3) is of relative size $(\tau_e/T)^n = \epsilon^n$. The first neglected term is therefore of order ϵ^2 , and equation (4.5) predicts a fractional acceleration-level difference scaling as ϵ^2 , except near zeros of the force where the denominator vanishes and relative errors become ill-conditioned.

This estimate also explains why the comparison is about time variation rather than force strength. In the linear Abraham–Lorentz benchmark, the amplitude of F multiplies every term in equation (4.3). It therefore cancels from any normalized difference between the physical acceleration and the LL acceleration. What remains is the rate at which the force changes over the short memory time τ_e . A strong but slowly varying field can still be well approximated by LL as a classical reduction of order, while a weak field that changes over a time comparable to τ_e would be a poor case for the first-order truncation. This point is artificial in the sense that realistic optical pulses never approach $T \sim \tau_e$, but it is useful because it isolates the mathematical small parameter of the reduction of order.

The estimate should be read as an error statement for an acceleration profile, not as a statement about total radiated energy alone. Energy loss depends on integrals of the acceleration and can sometimes hide local discrepancies through cancellations. By contrast, the acceleration comparison used here asks a stricter question: at each time, does the local LL rule reproduce the physical non-runaway acceleration selected by the future boundary condition? The answer is yes to first order in ϵ and no at second order, with the leading miss proportional to the curvature of the applied force. For the Gaussian pulse, this places the largest discrepancy near the pulse centre, where the force profile bends most strongly, rather than in the far tails where the force is already negligible.

The expansion must not be read as a proof that adding further reduction-of-order corrections always improves the approximation. The derivative series (4.3) is generally asymptotic rather than convergent: successive terms can grow, and the formal sum need not converge to the physical solution [9, 10]. In the constant-crossed-field case the reduction-of-order series is explicitly divergent, and resumming it reproduces a runaway-free physical solution rather than diverging without meaning [9]. The practical reading is that LL is the first, and often an optimal low-order, truncation. Equation (4.5) is a controlled leading estimate of its error for smooth pulses with $\epsilon \ll 1$, not an exact remainder, and it says nothing reliable about the behaviour of higher truncations.

There is a useful physical picture behind the same algebra. The exact branch in equation (3.4) averages the future force over an exponential window of width τ_e . LL replaces that weighted average by the value of the force now plus its first local slope. If the force is nearly straight over the memory window, this linear replacement is accurate. If the force has appreciable curvature over that window,

LL cannot know it, because all information beyond the first derivative has been discarded. The leading curvature term $\tau_e^2 \ddot{F}/m$ is therefore the mathematical trace of the nonlocal information lost when the physical LAD branch is compressed into a local second-order equation. This is why the benchmark uses a smooth pulse: it lets the local expansion fail gradually and measurably, rather than through a discontinuity where the derivative series itself would not be defined.

4.3 Hypotheses for the numerical section

The leading estimate (4.5) turns into three concrete predictions for the Gaussian-pulse comparison in Section 5.

1. A normalized root-mean-square difference between a_{LAD} and a_{LL} should scale approximately as ϵ^2 for $\epsilon \ll 1$, that is, with slope close to 2 on a log-log plot against ϵ .
2. The largest pointwise deviations should appear where $|\ddot{F}|$ is largest, near the pulse peak, and should grow as ϵ approaches unity, where the truncation (4.4) loses control.
3. Changing the field amplitude rescales a_{LAD} and a_{LL} together, so it changes χ_e but leaves the normalized LL-LAD error unchanged in this linear benchmark.

The next section tests all three against a direct numerical evaluation of the exact integral.

5 Gaussian-pulse numerical illustration

5.1 Model and dimensionless variables

The external field is a spatially uniform Gaussian electric pulse,

$$E(t) = E_0 \exp\left(-\frac{t^2}{2T^2}\right), \quad B = 0, \quad (5.1)$$

acting on an initially nonrelativistic electron, so that $F(t) = -eE(t)$ with peak magnitude $F_0 = eE_0$. The Gaussian is a convenient test profile because it is smooth, has a single time scale T , and has derivatives of every order, so the expansion of Section 4 applies cleanly. Three dimensionless variables organize the comparison,

$$s = \frac{t}{T}, \quad \epsilon = \frac{\tau_e}{T}, \quad \kappa = \frac{E_0}{E_S}. \quad (5.2)$$

Here s is time in units of the pulse duration, ϵ is the reduction-of-order parameter, and κ is the field amplitude in units of the Schwinger field. In the nonrelativistic electric-only limit, the quantum parameter of Section 6 is $\chi_e \approx \gamma\kappa$. Since the benchmark electron is nonrelativistic ($\gamma \approx 1$), $\chi_e \approx \kappa$ here, and the general $\gamma\kappa$ form is kept for Section 6. Thus κ sets χ_e while ϵ is fixed entirely by the pulse duration. The acceleration is reported in units of F_0/m , with an overall sign absorbed so that the normalized profiles are positive and the error metrics below are sign-independent ratios. In these units the normalized LL-LAD error depends only on ϵ , not on E_0 : rescaling the amplitude multiplies a_{LAD} and a_{LL} by the same factor. This is hypothesis 3 of Section 4.3, and it holds by construction in the linear model.

5.2 Numerical method and error metrics

In dimensionless form the exact physical acceleration follows from equation (3.4),

$$\frac{m a_{\text{LAD}}(s)}{F_0} = \int_0^\infty e^{-u} \exp \left[-\frac{(s + \epsilon u)^2}{2} \right] du, \quad (5.3)$$

and the LL approximation (4.4) becomes

$$\frac{m a_{\text{LL}}(s)}{F_0} = \exp \left(-\frac{s^2}{2} \right) - \epsilon s \exp \left(-\frac{s^2}{2} \right). \quad (5.4)$$

The two profiles are compared with a relative root-mean-square and a relative maximum metric,

$$D_{\text{rms}} = \left[\frac{\int (a_{\text{LAD}} - a_{\text{LL}})^2 ds}{\int a_{\text{LAD}}^2 ds} \right]^{1/2}, \quad D_{\text{max}} = \frac{\max_s |a_{\text{LAD}} - a_{\text{LL}}|}{\max_s |a_{\text{LAD}}|}. \quad (5.5)$$

The integral (5.3) is evaluated by composite Simpson quadrature in u . The kernel e^{-u} suppresses the integrand exponentially, so truncating the u range at $u_{\text{max}} = 40$ with 1200 subintervals reproduces the integral to far better than the precision reported here. The acceleration profiles are sampled on a uniform s grid over $[-8, 8]$, and the metric integrals in equation (5.5) are evaluated by the trapezoidal rule on that grid. The scan runs over $\epsilon \in [0.01, 1.0]$. The full calculation is performed by a short, dependency-free script using only the Python standard library, so the figures and Table 1 are reproducible from the source provided with this project.

The method needs little numerical machinery at the accuracy required here. The exponential weight makes the u integral effectively finite, and a convergence rerun with $u_{\text{max}} = 80$ and 2400 Simpson subintervals changed D_{rms} by less than 4×10^{-18} across the scan. Doubling the s grid changed D_{rms} by less than 2×10^{-11} and D_{max} by less than 1.5×10^{-5} , below the precision shown in the table. The one place where care is needed is the definition of the relative error itself. A pointwise ratio $|a_{\text{LAD}} - a_{\text{LL}}|/|a_{\text{LAD}}|$ diverges wherever a_{LAD} passes through zero, which the Gaussian acceleration does in its tails. The metrics in equation (5.5) are therefore built from integrated and peak-normalized quantities rather than from a pointwise ratio. This keeps D_{rms} and D_{max} finite and dominated by the region near the pulse where the comparison is physically meaningful.

The numerical calculation is intentionally more transparent than efficient. Since the problem is one-dimensional and the kernel is positive, direct quadrature is clearer than solving the differential equation as an initial-value problem and then trying to suppress its runaway mode. It also makes the physical boundary condition visible in the code: every value of $a_{\text{LAD}}(s)$ is obtained by averaging future values of the force with the weight e^{-u} . The script therefore implements the same object as the analytic derivation, not a different numerical regularization of LAD. This matters pedagogically, because the goal is to compare LL with the physical branch rather than with a numerically stabilized third-order trajectory.

The scan separates three kinds of evidence. Table 1 tests the scaling law quantitatively, Figure 1 shows the local shape of the discrepancy, and Figure 2 checks whether the same scaling survives across several decades of ϵ . No single one of these is sufficient. A table alone could hide where in the pulse the error occurs; profile plots alone could exaggerate a visually striking but small integrated error; and a log-log plot alone would not fix the numerical coefficient. Taken together they provide a compact consistency check on the analytic argument.

5.3 Results

Table 1 lists the two error metrics over the scanned range, together with the ratio $D_{\text{rms}}/\epsilon^2$. This ratio is the cleanest test of the predicted scaling: if the leading error is $O(\epsilon^2)$, it should approach a constant as $\epsilon \rightarrow 0$. It does: the value settles near 0.86 for the smallest pulses and drifts downward only as ϵ grows toward unity, where the higher-order terms dropped by LL begin to matter.

Table 1. Numerical comparison for the Gaussian pulse, from direct quadrature of equation (5.3) with the documented script. The near-constant ratio $D_{\text{rms}}/\epsilon^2$ at small ϵ confirms the predicted leading scaling; its decline at large ϵ marks the onset of higher-order corrections.

$\epsilon = \tau_c/T$	D_{rms}	D_{max}	$D_{\text{rms}}/\epsilon^2$
0.02	3.46×10^{-4}	4.00×10^{-4}	0.866
0.05	2.16×10^{-3}	2.49×10^{-3}	0.864
0.10	8.58×10^{-3}	9.90×10^{-3}	0.858
0.20	3.34×10^{-2}	3.86×10^{-2}	0.835
0.50	1.83×10^{-1}	2.14×10^{-1}	0.732
1.00	5.83×10^{-1}	7.08×10^{-1}	0.583

Figure 1 shows the acceleration profiles for a smooth pulse and a short pulse. At $\epsilon = 0.05$, the exact physical acceleration and the LL approximation are visually indistinguishable across the whole pulse. At $\epsilon = 0.50$, they separate: LL overshoots the peak and is displaced in time relative to the exact curve, and the shaded difference is a sizeable fraction of the signal. The departure is concentrated near the pulse peak, where $|\ddot{F}|$ is largest, as anticipated by hypothesis 2.

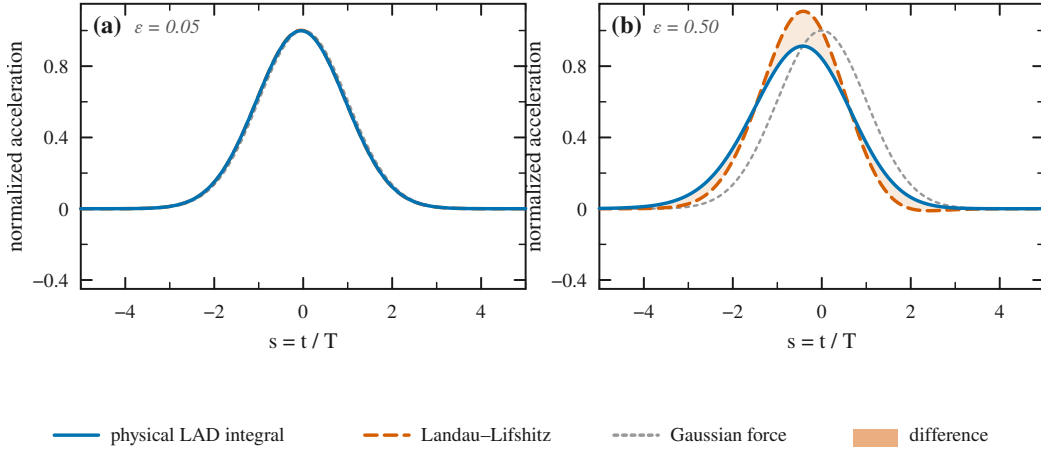


Figure 1. Normalized acceleration for a smooth pulse ($\epsilon = 0.05$, panel a) and a short pulse ($\epsilon = 0.50$, panel b). The exact physical LAD integral (5.3) (blue) is compared with the LL approximation (5.4) (vermillion) and the driving Gaussian force (grey). The shaded band marks their difference, which is negligible at $\epsilon = 0.05$ and pronounced at $\epsilon = 0.50$.

Figure 2 plots the two metrics against ϵ on logarithmic axes. For $\epsilon \lesssim 0.1$, both D_{rms} and D_{max} track the quadratic guide line of slope 2, confirming hypothesis 1. As ϵ approaches unity, the curves bend below slope 2. This is consistent with the decline of $D_{\text{rms}}/\epsilon^2$ in Table 1: once the pulse is no

longer slow on the scale τ_e , the single ϵ^2 term ceases to capture the full error and the comparison enters the regime where LL is not a controlled approximation.

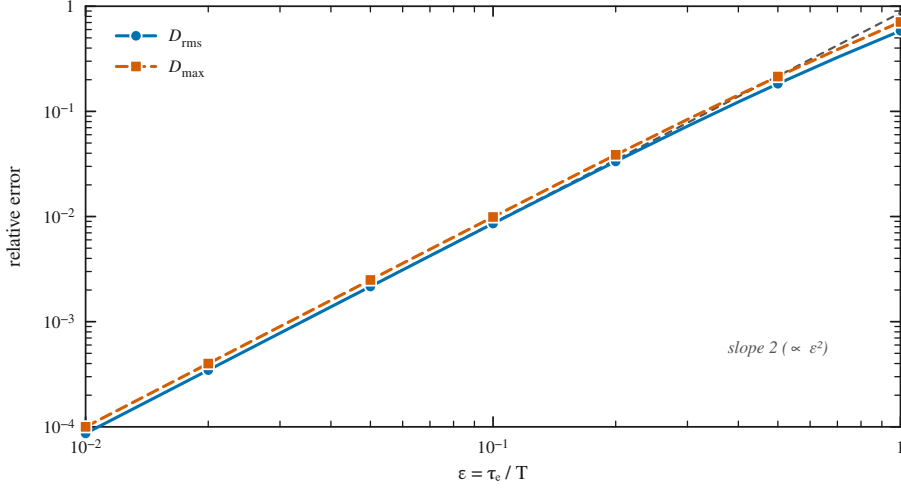


Figure 2. Relative error metrics D_{rms} (blue circles) and D_{max} (vermillion squares) versus $\epsilon = \tau_e/T$ on log–log axes, with a slope-2 guide. The small- ϵ behaviour matches the predicted ϵ^2 scaling; the roll-off near $\epsilon \sim 1$ signals the breakdown of the first-order truncation.

The comparison also displays the time structure anticipated from the future-weighted integral. Both accelerations peak slightly before the force does, at $s < 0$, because the kernel in equation (5.3) samples the force ahead of the current instant. This is the pre-acceleration of Section 3 made visible. The LL peak position is itself analytic: setting the derivative of equation (5.4) to zero gives

$$s_{\text{peak}}^{\text{LL}} = \frac{1 - \sqrt{1 + 4\epsilon^2}}{2\epsilon} \approx -\epsilon, \quad (5.6)$$

so the LL acceleration peaks a time of order ϵ (in units of T) ahead of the force. At $\epsilon = 0.5$ this evaluates to $s_{\text{peak}}^{\text{LL}} \approx -0.41$, in agreement with the tabulated value, and the exact peak sits a little earlier still. At small ϵ the exact and LL peaks coincide, but as ϵ grows the LL peak lags the exact one: the positions stay within a grid spacing of each other up to $\epsilon \approx 0.3$ and then separate, reaching a displacement of order 0.08 in s at $\epsilon = 1$. The peak-height error and this peak displacement together account for the visible mismatch in Figure 1(b).

These results confirm all three hypotheses of Section 4.3: the error scales as ϵ^2 at small ϵ , it is largest near the pulse peak and grows toward $\epsilon \sim 1$, and it is independent of the field amplitude in the normalized linear model. The large- ϵ entries should be read as diagnostics of the approximation rather than descriptions of realistic pulses, since $\epsilon \sim 1$ would require a pulse duration comparable to $\tau_e \sim 6 \times 10^{-24}$ s.

The independence from amplitude is important because it connects the numerical section to the discussion of χ_e . If the pulse is multiplied by a constant factor, the profiles in Figure 1 grow by that same factor before normalization and the metrics in Table 1 are unchanged. Yet the same amplitude change would raise the rest-frame field and hence raise χ_e . The Gaussian benchmark

therefore demonstrates, rather than merely asserts, that the classical LL–LAD error measured here is not a hidden quantum parameter in disguise. The field amplitude can make the classical model physically questionable without changing how well LL tracks the physical classical branch in this linear limit; this is why the regime map in the next section can display the computed error as vertical contours in the (ϵ, χ_e) plane.

5.4 Analytic check of the leading coefficient

Closed-form Gaussian accelerations have been used before to compare Abraham–Lorentz and LL motion [7]. The added step here is to convert that comparison into a normalized error benchmark with a known small- ϵ coefficient.

The small- ϵ behaviour can be predicted in closed form, which provides an independent check on the quadrature. In the normalized units of equation (5.3) the leading error from equation (4.5) is

$$\frac{m}{F_0} (a_{\text{LAD}} - a_{\text{LL}}) \simeq \epsilon^2 g''(s), \quad g(s) = e^{-s^2/2}, \quad g''(s) = (s^2 - 1) e^{-s^2/2}, \quad (5.7)$$

since each time derivative brings a factor $1/T$ and the radiation-reaction prefactor τ_e^2 supplies the ϵ^2 . At small ϵ both accelerations reduce to $g(s)$, so the denominators in equation (5.5) are Gaussian integrals. Using $\int e^{-s^2} ds = \sqrt{\pi}$, $\int s^2 e^{-s^2} ds = \sqrt{\pi}/2$, and $\int s^4 e^{-s^2} ds = 3\sqrt{\pi}/4$, the root-mean-square metric becomes

$$D_{\text{rms}} \xrightarrow{\epsilon \rightarrow 0} \epsilon^2 \left[\frac{\int (s^2 - 1)^2 e^{-s^2} ds}{\int e^{-s^2} ds} \right]^{1/2} = \epsilon^2 \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2} \epsilon^2 \approx 0.866 \epsilon^2, \quad (5.8)$$

while the maximum metric is set by the peak of $|g''|$, which occurs at $s = 0$ with $|g''(0)| = 1 = g(0)$, giving $D_{\text{max}} \rightarrow \epsilon^2$. These two predictions, $D_{\text{rms}}/\epsilon^2 \rightarrow \sqrt{3}/2$ and $D_{\text{max}}/\epsilon^2 \rightarrow 1$, match the smallest- ϵ rows of Table 1 to the quoted precision, confirming both the ϵ^2 scaling and the absolute normalization of the numerical comparison.

The same expansion predicts how the ratio $D_{\text{rms}}/\epsilon^2$ should drift away from $\sqrt{3}/2$ as ϵ grows, which is a more demanding test than the leading limit alone. Carrying the series (4.3) one order further and keeping the next non-vanishing contribution to each integral in equation (5.5) gives

$$\frac{D_{\text{rms}}}{\epsilon^2} = \frac{\sqrt{3}}{2} [1 - \epsilon^2 + O(\epsilon^4)], \quad (5.9)$$

where the unit coefficient of ϵ^2 follows from the Gaussian-moment ratios $\int g'^2 / \int g^2 = \frac{1}{2}$ and $\int g'''^2 / \int g''^2 = \frac{5}{2}$, with the odd cross terms integrating to zero by symmetry. Equation (5.9) predicts $D_{\text{rms}}/\epsilon^2 = 0.866, 0.857, 0.831$ at $\epsilon = 0.05, 0.1, 0.2$, against the tabulated 0.864, 0.858, 0.835. The agreement holds until the residual $O(\epsilon^4)$ becomes visible beyond $\epsilon \sim 0.3$. The downward drift of the ratio in Table 1 is therefore not numerical scatter but the first higher-order reduction-of-order term making itself felt, the same term whose neglect defines the LL truncation. This closes the loop between the analytic argument of Section 4 and the numerics: the leading coefficient, its first correction, and the onset of the roll-off are all accounted for. The two metrics keep a nearly constant ratio across the scan, $D_{\text{max}}/D_{\text{rms}} \approx 1.15$ to 1.21, which reflects that the error retains a fixed shape, peaked at the pulse centre, while its overall size grows with ϵ ; the slow rise of the ratio toward large ϵ is the shape beginning to distort as higher-order terms enter.

With ϵ established as the parameter that governs the classical reduction-of-order error, the remaining question is how this criterion sits beside the quantum parameter χ_e , which is the subject of Section 6.

6 Relation to χ_e and strong-field radiation reaction

6.1 Definition and physical meaning of χ_e

The benchmark of the previous sections answers a purely classical question. To place that answer in its physical context, it must be compared with the parameter that governs when a classical description of radiation is expected to hold at all. The strong-field quantum parameter for an electron is

$$\chi_e = \frac{e\hbar}{m^3 c^4} \sqrt{-(F^{\mu\nu} p_\nu)^2} \simeq \frac{\gamma E_{\text{eff}}}{E_S}, \quad E_S = \frac{m^2 c^3}{e\hbar} \approx 1.3 \times 10^{18} \text{ V m}^{-1}, \quad (6.1)$$

where E_{eff} is the field experienced by the electron and E_S is the Schwinger field, the field at which the work done over a reduced Compton wavelength reaches the electron rest energy.¹ Physically, χ_e compares the field strength in the electron rest frame with the Schwinger scale and controls the photon recoil per emission.

The covariant form in equation (6.1) makes this frame-independence explicit. The combination $\sqrt{-(F^{\mu\nu} p_\nu)^2}$ is the magnitude of the electromagnetic four-force on the electron, which in the rest frame reduces to the local field strength. Dividing by E_S expresses that rest-frame field in the units at which quantum electrodynamics becomes strongly nonlinear and the vacuum can break down through pair production. A value $\chi_e \sim 1$ therefore signals that the field seen by the electron is approaching the scale at which a single emitted photon can carry away a finite fraction of the electron energy. At that point, the continuous and deterministic picture of radiation loss underlying any classical equation, including LL, loses its footing.

The size of χ_e sets which description of emission is adequate. For $\chi_e \ll 1$, the recoil from each emitted photon is negligible, radiation is well approximated as a continuous classical energy loss, and a deterministic classical radiation-reaction equation can be meaningful. Quantum corrections to the emission spectrum become necessary around $\chi_e \gtrsim 0.1$, and for $\chi_e \gtrsim 1$ the recoil per photon is large and emission is strongly stochastic; the hard photons produced in that regime can also seed pair-production channels. In that regime, a disagreement between LL and experiment is not evidence that the LL reduction of LAD is a poor classical approximation. It is evidence that the classical model itself has been pushed past its domain, or that the experimental modelling includes effects outside this toy benchmark. Recent strong-field experiments report exactly this transition, with data favouring quantum-continuous and quantum-stochastic descriptions over the classical one as χ_e approaches and exceeds unity [4, 12–14].

A rough estimate fixes the scale. For a head-on, counterpropagating collision, a relativistic electron sees a rest-frame field of order $2\gamma E_L$, where the factor of two is the relativistic Doppler enhancement. Thus $\chi_e \sim 2\gamma E_L/E_S$. Taking a gigaelectronvolt-scale electron, $\gamma \sim 2 \times 10^3$, against a pulse of intensity $\sim 10^{21} \text{ W cm}^{-2}$, whose lab field is $E_L \sim 10^{14} \text{ V m}^{-1}$, gives χ_e of a few tenths.

¹Notation varies in the literature. Some authors write η for the quantum parameter called χ_e here; in the convention common to strong-field QED experiments η denotes instead the energy parameter and the quantum nonlinearity parameter is $\chi = \xi\eta$, with $\xi = a_0$ the classical nonlinearity parameter. This paper uses χ_e throughout.

Pushing χ_e to order unity requires higher electron energy, higher intensity, or both, which is the direction current facilities are taking. These numbers matter here only because they show where laboratory work sits relative to the classical-accuracy axis, a point made precise in Section 6.4.

6.2 Why ϵ and χ_e are not the same parameter

The benchmark makes the distinction between the two parameters concrete. The reduction-of-order error studied in Sections 4 and 5 is governed by $\epsilon = \tau_e/T$, which measures how fast the field varies in units of the radiation-reaction time. The quantum parameter $\chi_e \simeq \gamma\kappa$ is instead an amplitude-and-energy measure, set by the field strength and the electron energy. In the normalized linear benchmark these are independent by construction: changing the field amplitude E_0 changes κ and hence χ_e , but leaves the normalized D_{rms} untouched, because a_{LAD} and a_{LL} rescale together. The pulse duration T controls ϵ ; the amplitude controls χ_e ; the two knobs are separate.

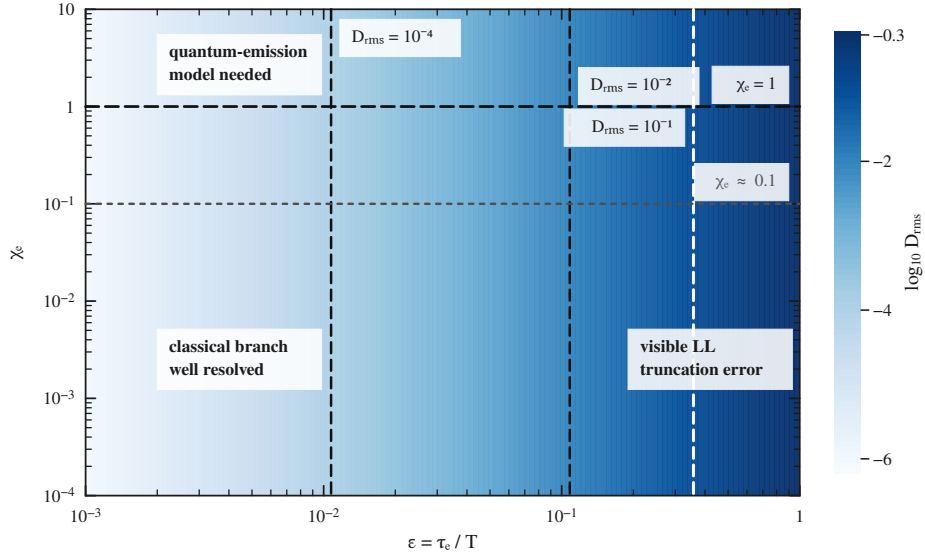
This separation is a property of the model, not a universal decoupling. In realistic relativistic problems, high γ and high field tend to appear together, so ϵ -type and χ_e -type effects are correlated in practice. The exact solution of the LL equation in a plane wave shows how rich the classical relativistic dynamics already is [8], and a detailed analysis of a relativistic electron in a counter-propagating laser finds that LL becomes an inaccurate approximation to LAD in regimes where quantum recoil is also significant [11]. The toy model is valuable precisely because it holds the two apart and shows that the correlation is contingent, not an identity.

The mechanism behind that correlation is worth stating. In a plane wave, the field variation actually experienced by the electron is set by the laser phase, Doppler-compressed by the electron's own counterpropagating motion. The effective rate of variation therefore carries factors of γ and of the field strength, the same factors that build χ_e . The clean factorization of the nonrelativistic benchmark, where the duration T and the amplitude E_0 are independent dials, is exactly what relativistic kinematics removes: there is no longer a knob that changes the variation scale without also changing the rest-frame field. The benchmark is thus the controlled limit in which the two effects can be seen to be distinct in principle, even though no single laboratory parameter isolates them.

A third axis is useful for keeping the logic clean. The strength of radiation reaction itself, as opposed to the accuracy of its reduced form or the validity of its classical description, is measured by a classical radiation-dominance parameter $R_c = \alpha a_0 \chi_e$, which becomes of order unity when radiative energy loss strongly alters the trajectory. The present benchmark deliberately keeps R_c small: the motion is nonrelativistic and the radiation-reaction term is a small perturbation, which is why ϵ and χ_e read as clean orthogonal axes. Realistic strong-field experiments raise a_0 , χ_e , and R_c together, which is the physical reason the three questions become entangled there even though they are logically distinct.

6.3 Diagnostic error map in the (ϵ, χ_e) plane

Combining the classical-accuracy axis ϵ with the quantum-validity axis χ_e gives the diagnostic picture in Figure 3. Unlike a quadrant schematic, the colour field and vertical contours encode the computed Gaussian-benchmark error $D_{\text{rms}}(\epsilon)$. The horizontal guides remain interpretive strong-field markers: $\chi_e \sim 0.1$, where quantum corrections to the spectrum begin to matter, and $\chi_e \sim 1$, where recoil and stochastic emission become strong.



Vertical contours come from the Gaussian benchmark; horizontal guides are χ_e thresholds.

Figure 3. Diagnostic error map in the (ϵ, χ_e) plane. The colour scale and vertical contours encode the computed Gaussian-benchmark error $D_{\text{rms}}(\epsilon)$, which is independent of χ_e in the normalized linear model. The horizontal guides mark $\chi_e \approx 0.1$, where quantum spectral corrections begin to matter, and $\chi_e = 1$, where recoil and stochastic emission become strong. The map is therefore a diagnostic of which question is being asked: horizontal motion changes the LL reduction-of-order error, while vertical motion changes the physical validity of a classical emission model.

The operational reading is simple. At small ϵ and small χ_e , LL is both a controlled reduction of order and a physically adequate classical model. At large ϵ but small χ_e , the repair is a higher-order or resummed classical treatment. At small ϵ but large χ_e , LL may still approximate the physical classical branch, but that branch itself needs quantum-emission physics. When both parameters are large, neither LL nor a purely classical LAD branch is sufficient.

The map in Figure 3 is the paper’s strongest positioning contribution: it turns the qualitative phrase “LL breaks down” into a quantitative diagnostic chart for this benchmark. Quoting a single value of χ_e diagnoses only the vertical axis; the benchmark was built to show why that is not the same as diagnosing the LL reduction of order.

6.4 Where laboratory experiments sit

It is instructive to locate real strong-field experiments on the map, because they occupy only a narrow part of it. Optical and near-infrared laser pulses have durations in the femtosecond range, $T \sim 10^{-15}$ to 10^{-14} s, while $\tau_e \sim 6 \times 10^{-24}$ s. The reduction-of-order parameter is therefore

$$\epsilon = \frac{\tau_e}{T} \sim 10^{-9}, \quad (6.2)$$

an extremely small number. For any realistic pulse, the LL truncation error isolated in this benchmark is negligible, of order $\epsilon^2 \sim 10^{-18}$, and is far below any laboratory sensitivity. Laboratory experiments therefore live at the extreme left edge of Figure 3 for the specific source of classical error studied here.

What those experiments vary is the vertical axis. The collision measurements of Cole et al. and Poder et al. reached χ_e of order a few tenths, and the recent observation of Los et al. probed the regime where χ_e approaches unity and quantum models are favoured over the classical one [12–14]. On the map, they trace an almost purely vertical path, climbing the χ_e axis from the safe corner at fixed, negligible ϵ . The consequence is sharp but limited: a measurable disagreement between an LL-based classical prediction and a real strong-field experiment in this laboratory regime would not be plausibly explained by the reduction-of-order truncation isolated here. It would point instead to quantum emission, beam and laser modelling, or detector/background effects. The large- ϵ region explored numerically in Section 5 is unreachable in the laboratory; its role is to expose the classical reduction-of-order structure that the χ_e axis would otherwise hide.

This reading also settles the modelling choice raised in Section 1. Because every realistic pulse sits at $\epsilon \sim 10^{-9}$, the reduction-of-order error is not the limiting factor in a laboratory simulation. The relevant decision is mainly vertical: a deterministic classical or LL force while $\chi_e \ll 1$, a quantum-continuous correction as χ_e approaches 0.1, and a stochastic emission treatment once $\chi_e \sim 1$. A code that selects its radiation-reaction model on the value of χ_e alone is therefore well justified in the laboratory regime, not by accident but because ϵ is fixed and negligible there. The benchmark does not overturn that practice; it explains why it is sound, and it identifies the unphysical corner of parameter space, short pulses with ϵ of order unity, where the justification would break down.

6.5 Outlook: the relativistic regime

The benchmark is built to hold ϵ and χ_e apart. The natural next question is how they recombine once that separation is relaxed. A relativistic one-dimensional extension would replace the prescribed nonrelativistic force by the full Lorentz force of a strong field and the Abraham–Lorentz term by the covariant self-force of equation (2.1). The field variation seen by the electron would then no longer be set by a single external time scale T but by the Doppler-compressed phase, which depends on the electron’s own motion. The effective reduction-of-order parameter would become a function of γ and of the field strength, and the clean horizontal axis of Figure 3 would tilt.

The plane-wave literature already indicates the shape of that tilt. Exact LL solutions exist for arbitrary plane waves [8], and the accuracy criteria derived for a relativistic electron in a counterpropagating laser place the LL–LAD discrepancy in the same strong-field corner where quantum recoil sets in [11]. A natural continuation of the present work would compute the analogue of D_{rms} along a physical trajectory in such a field and track how the small- ϵ scaling deforms as χ_e grows, extending the present diagnostic map into the relativistic regime. That calculation lies beyond the classical, nonrelativistic scope adopted here, but the benchmark identifies precisely the quantity it should measure and the axis along which the deformation should be reported.

7 Limitations and conclusion

7.1 Limitations

The benchmark is deliberately narrow, and its limits should be stated plainly. The model is nonrelativistic, one-dimensional, and driven by an externally prescribed field. It captures the

reduction-of-order structure, but not the geometry of a real laser–electron collision, where the field depends on position, the magnetic force matters, and the electron is relativistic. The field is treated as a fixed background; back-reaction on the field and collective effects are absent by construction.

A second limitation is dimensional. Because $\tau_e \sim 6 \times 10^{-24}$ s is so small, a pulse with T comparable to τ_e is far outside any realistic laboratory regime, and at such scales the point-particle classical model is itself questionable. The large- ϵ entries in Section 5 are therefore diagnostics of the approximation, chosen to expose the scaling, not predictions for an experiment. Finally, the paper makes no attempt at a relativistic LAD integration, an exact plane-wave treatment, or a stochastic quantum-emission calculation. The strong-field discussion of Section 6 is interpretive, and the quantitative claims are confined to the classical toy model.

A further caveat concerns the point-particle idealization. The Abraham–Lorentz model gives the electron no internal structure, uses a future boundary condition to remove the runaway mode, and omits spin. These are serious limitations if the result is interpreted as a literal electron trajectory, but they do not change the benchmark’s purpose: testing the local LL truncation against an exactly specified physical branch.

The simplicity of the benchmark also means that it should not be used to rank competing radiation-reaction models in realistic simulations. In the present calculation the external force is prescribed, so the electron trajectory does not feed back into the force it samples. A real laser pulse changes along the particle path, and the particle path changes because of radiation reaction. That feedback is where the relativistic LL equation earns its usefulness and where a direct comparison with LAD becomes technically difficult. The present result is therefore best read as a calibration calculation: it fixes the meaning of “LL approximates the physical LAD branch” in a case where the answer is unambiguous, and it supplies a small parameter that can be looked for, in modified form, in more complicated problems.

There is also a limitation in the way the quantum discussion is framed. The parameter χ_e is a powerful diagnostic, but it is not a complete theory of strong-field emission. It summarizes the typical recoil scale and helps decide when quantum spectra and stochastic emission matter, yet detailed predictions require an emission model, particle distribution, laser geometry, and detector response. The two-axis map in Section 6 should therefore be understood as a conceptual classification, not as a replacement for a full strong-field QED or Monte Carlo calculation. Its value is to prevent a category error: using χ_e to diagnose the quantum validity of the classical picture should not be confused with diagnosing the mathematical accuracy of the LL reduction of order.

7.2 Conclusion

In the nonrelativistic one-dimensional Abraham–Lorentz limit of the LAD equation, the physical non-runaway acceleration is an exact future-weighted integral of the applied force, and the Landau–Lifshitz acceleration is its first local reduction-of-order approximation. For a smooth Gaussian pulse, the difference between the two is controlled by the single parameter $\epsilon = \tau_e/T$: the leading local error is of order ϵ^2 , and the numerical comparison confirms a root-mean-square difference that scales as ϵ^2 for $\epsilon \ll 1$ and grows as ϵ approaches unity, where the first-order truncation ceases to be controlled. This is the direct answer to the first half of the research question.

The second half concerns χ_e . The quantum nonlinearity parameter answers a different question from ϵ : not whether LL approximates the physical classical branch, but whether that classical branch

is a physically adequate model at all. In the linear benchmark the two are independent, since the field amplitude sets χ_e but cancels from the normalized LL–LAD error. A third parameter, the radiation-dominance R_c , measures the strength of radiation reaction and is held small here. The diagnostic error map of Section 6 turns this into a usable picture: reduction-of-order breakdown is a move along the ϵ axis and is repaired by a higher-order or resummed classical treatment, whereas quantum-validity breakdown is a move along the χ_e axis and requires a quantum-emission description. A single value of χ_e diagnoses only the second axis.

This conclusion has a practical modelling consequence. In ordinary laser experiments, ϵ is so tiny that LL and the physical classical branch are indistinguishable for the benchmarked source of error. The important modelling decision is therefore not LAD versus LL but classical continuous loss versus quantum emission. Conversely, an artificial short-pulse benchmark with large ϵ can make LL visibly fail while still having small χ_e , in which case replacing LL by a stochastic quantum model would answer the wrong question. The correct repair depends on the direction of failure in the diagnostic map.

The most defensible application is therefore not nucleon structure or a direct BSM signal, but simulation hygiene for accelerator-based strong-field QED. Experiments such as LUXE are designed to measure electron, positron and photon spectra over large flux ranges, and their design relies on detector simulations and strong-field event modelling [17]. QED-PIC calculations similarly switch between deterministic radiation-reaction forces and stochastic photon-emission modules according to the separation between classical recoil and quantum emission probabilities [15, 16]. Future lepton-collider studies add a related particle-physics context: beamstrahlung and beam-induced backgrounds are characterized in part by a collider strong-field parameter, often denoted Y , and affect luminosity spectra, pair backgrounds and detector occupancies [18, 19]. The Gaussian benchmark is too simple to predict those systems, but it supplies a portable diagnostic question: is a discrepancy a reduction-of-order error, a quantum-emission effect, or a detector/background-modelling issue?

Against existing Abraham–Lorentz/LL comparisons, the contribution is not the Gaussian solution alone but the calibrated error statement, the reproducible benchmark workflow, and the diagnostic error map that separates the classical parameter ϵ from the quantum parameter χ_e . The benchmark, the ϵ^2 error law, and the two-parameter diagnostic map separate two notions of breakdown that the strong-field literature often discusses together, clarifying which remedy a given regime calls for.

A Gaussian moment integrals for the error coefficients

The leading and next-to-leading coefficients quoted in Section 5.4 follow from a compact set of Gaussian integrals, collected here for completeness. Writing $g(s) = e^{-s^2/2}$, the derivatives that enter the error are

$$g' = -s g, \quad g'' = (s^2 - 1) g, \quad g''' = s(3 - s^2) g, \quad g'''' = (s^4 - 6s^2 + 3) g. \quad (\text{A.1})$$

Every integral needed reduces to the standard even Gaussian moments

$$\int_{-\infty}^{\infty} s^{2k} e^{-s^2} ds = \frac{(2k-1)!!}{2^k} \sqrt{\pi}, \quad (\text{A.2})$$

giving $\int e^{-s^2} = \sqrt{\pi}$, $\int s^2 e^{-s^2} = \frac{1}{2}\sqrt{\pi}$, $\int s^4 e^{-s^2} = \frac{3}{4}\sqrt{\pi}$, and $\int s^6 e^{-s^2} = \frac{15}{8}\sqrt{\pi}$. Substituting the squares of equation (A.1),

$$\int g^2 = \sqrt{\pi}, \quad \int g'^2 = \frac{\sqrt{\pi}}{2}, \quad \int g''^2 = \frac{3\sqrt{\pi}}{4}, \quad \int g'''^2 = \frac{15\sqrt{\pi}}{8}, \quad (\text{A.3})$$

where all integrals run over the real line.

At leading order in ϵ the normalized error is $\epsilon^2 g''$ and the denominators in equation (5.5) reduce to $\int g^2$, so

$$\frac{D_{\text{rms}}}{\epsilon^2} \Big|_{\epsilon \rightarrow 0} = \left(\frac{\int g''^2}{\int g^2} \right)^{1/2} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}, \quad \frac{D_{\text{max}}}{\epsilon^2} \Big|_{\epsilon \rightarrow 0} = \frac{|g''(0)|}{g(0)} = 1, \quad (\text{A.4})$$

since $|g''|$ attains its maximum value 1 at $s = 0$, where $g = 1$. For the next order, the numerator and denominator of D_{rms}^2 pick up the first corrections from the higher terms of the series (4.3); using $\int g'' g'''' = -\int g'''^2$ (integration by parts), with the odd cross terms vanishing by parity,

$$N = \epsilon^4 \int g''^2 - \epsilon^6 \int g'''^2 + \dots, \quad D = \int g^2 - \epsilon^2 \int g'^2 + \dots. \quad (\text{A.5})$$

Forming the ratio and expanding,

$$\frac{D_{\text{rms}}^2}{\epsilon^4} = \frac{\int g''^2}{\int g^2} \left(1 - \epsilon^2 \frac{\int g'''^2}{\int g''^2} \right) \left(1 - \epsilon^2 \frac{\int g'^2}{\int g^2} \right)^{-1} + \dots = \frac{3}{4} (1 - 2\epsilon^2) + \dots, \quad (\text{A.6})$$

where $\int g'''^2 / \int g''^2 = \frac{5}{2}$ and $\int g'^2 / \int g^2 = \frac{1}{2}$. Taking the square root gives equation (5.9), $D_{\text{rms}}/\epsilon^2 = (\sqrt{3}/2)(1 - \epsilon^2 + \dots)$, which is the curve compared with Table 1 in Section 5.4.

B Dimensionless form of the benchmark

The dimensionless expressions (5.3) and (5.4) follow from the closed forms of Section 3 by direct substitution. For the Gaussian pulse the force is $F(t) = -F_0 e^{-t^2/2T^2}$ with $F_0 = eE_0$, and with $s = t/T$ and $\epsilon = \tau_e/T$ the shifted argument in equation (3.4) becomes $t + \tau_e u = T(s + \epsilon u)$. Hence

$$a_{\text{LAD}}(t) = \frac{1}{m} \int_0^\infty e^{-u} F(t + \tau_e u) du = -\frac{F_0}{m} \int_0^\infty e^{-u} \exp\left[-\frac{(s + \epsilon u)^2}{2}\right] du, \quad (\text{B.1})$$

and dividing by $-F_0/m$ (absorbing the overall sign as in Section 5.1) reproduces equation (5.3). For the LL acceleration, $\dot{F} = -F_0(d/dt)e^{-t^2/2T^2} = (F_0 s/T)e^{-s^2/2}$, so

$$a_{\text{LL}}(t) = \frac{1}{m} [F + \tau_e \dot{F}] = -\frac{F_0}{m} e^{-s^2/2} (1 - \epsilon s), \quad (\text{B.2})$$

which on dividing by $-F_0/m$ gives equation (5.4). The factor $1 - \epsilon s$ is the origin of the time displacement of the LL peak discussed in Section 5.3.

Acknowledgments

Data availability: all simulation artifacts are provided in `Simulations/`, including the script `run_gaussian_benchmark.py`, CSV data, table source, and figure files. Running the script reproduces Table 1 and Figures 1–3.

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